

Model Analysis Report

GaugeHD Synthetic

MKM-GHD-001

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Governance Framework: SR 11-7 / SS1/23 Model Risk Management

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1 Sensitivity Analysis

This section presents the sensitivity of model outputs to key input parameters. Parameter perturbation ranges are chosen to cover plausible operating conditions and stress scenarios as required under SR 11-7 guidance.

Table 1: Synthetic series volatility by AR(1) parameter ($\sigma_\epsilon = 1$).

ϕ (AR1)	Sample std dev	Theoretical variance
0.0000	0.9400	1.00
0.2000	1.05	1.04
0.4000	1.12	1.19
0.6000	1.37	1.56
0.8000	1.52	2.78
0.9000	2.42	5.26
0.9500	3.85	10.3

Table 2: Synthetic gauge range by seasonal amplitude (mean=2.0m, noise $\sigma = 0.5$).

Amplitude (m)	Min (m)	Max (m)	Range (m)
0.0000	0.5700	3.53	2.96
0.5000	0.5200	3.69	3.16
1.00	0.1300	4.00	3.86
2.00	-1.12	5.42	6.54
3.00	-2.27	6.10	8.37
5.00	-4.49	8.20	12.7

Table 3: Mean daily tidal range by tidal influence weight.

Tidal weight	Mean daily range (m)
0.0000	0.0000
0.1000	0.2000
0.2000	0.4000
0.3000	0.6000
0.5000	1.00
0.7000	1.40
1.00	2.00

2 Stress Testing

This section documents stress testing scenarios applied to the model under extreme but plausible conditions.

2.1 Stress Scenarios

Table 4: Stress testing scenarios for GaugeHD Synthetic.

Scenario	Parameter Shock	Severity	Status
Baseline	No shock	—	Passed
Moderate stress	+2 σ inputs	Medium	Pending
Severe stress	+3 σ inputs	High	Pending
Reverse stress	Output threshold breach	Critical	Pending

Detailed stress test results will be populated as scenarios are executed. Refer to the model's validation plan for the full stress testing programme.

3 Model Performance

This section tracks key performance metrics for the model across validation and production environments.

Table 5: Performance metrics for GaugeHD Synthetic.

Metric	Value	Status
Unit test pass rate	See Test Results	—
Execution time (single run)	TBD	—
Numerical stability	TBD	—
Reproducibility	Deterministic	OK

This report is produced automatically; human review is required before formal model approval. Supporting artefacts are available in the model documentation directory.